

8.2.4 Method of Steepest Descent ¶

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fit width



For a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ that we are trying to minimize, for a given x , the direction in which the function most rapidly increases in value at x is given by its gradient,

$$\nabla f(x).$$

Thus, the direction in which it decreases most rapidly is

$$-\nabla f(x).$$

For our function

$$f(x) = \frac{1}{2}x^T Ax - x^T b$$

this direction of steepest descent is given by

$$-\nabla f(x) = -(Ax - b) = b - Ax,$$

which we recognize as the residual. Thus, recalling that $r^{(k)} = b - Ax^{(k)}$, the direction of steepest descent at $x^{(k)}$ is given by $p^{(k)} = r^{(k)} = b - Ax^{(k)}$. These insights motivate the algorithms in [Figure 8.2.4.1](#).

Given : $A, b, x^{(0)}$	Given : A, b, x
$r^{(0)} := b - Ax^{(0)}$	$k := 0$
$k := 0$	$r := b - Ax$
while $r^{(k)} \neq 0$	while $r \neq 0$
$p^{(k)} := r^{(k)}$	$p := r$
$q^{(k)} := Ap^{(k)}$	$q := Ap$
$\alpha_k := \frac{p^{(k)T} r^{(k)}}{p^{(k)T} q^{(k)}}$	$\alpha := \frac{p^T r}{p^T q}$
$x^{(k+1)} := x^{(k)} + \alpha_k p^{(k)}$	$x := x + \alpha p$
$r^{(k+1)} := r^{(k)} - \alpha_k q^{(k)}$	$r := r - \alpha q$
$k := k + 1$	$k := k + 1$
endwhile	endwhile

Figure 8.2.4.1. Steepest descent algorithm, with indices and without indices.